role of bacteriophage in gene transfer, the lifecycle of bacteriophage is described below briefly.

After infecting the host cell, a bacteriophage (phage for short) often takes control and forces the host to make many copies of the virus. Eventually the host bacterium bursts or lyses and releases new phages. This reproductive cycle is called a lytic cycle because it ends in lysis of the host.

The lytic cycle (Figure 12.18) has four phases.

1. Attachment - Virus particle attaches to a specific receptor site on the bacterial surface.
2. Penetration - the genetic material, which is often double stranded DNA, then enters the cell.
3. Biosynthesis - After adsorption and penetration, the virus chromosome forces the bacterium to make viral components-viral nucleic acids and proteins.
4. Assembly - Phages are assembled from the virus components. Phage nucleic acid is packed within the virus's protein coat.
5. Release - mature viruses are released by cell lysis.

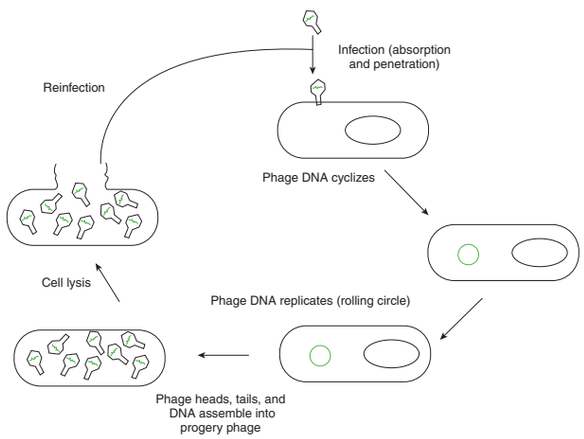


Figure 12.18: Lytic cycle

Bacterial viruses that reproduce using a lytic cycle often are called **virulent bacteriophages** (e.g. T phages) because they destroy the host cell. The genome of many DNA phages such as the lambda phage, after adsorption and penetration do not take control of its host and does not destroy the host. Instead the viral genome remains within the host cell and is reproduced along with the bacterial chromosome. The infected bacteria may multiply for long periods while appearing perfectly normal. Each of these infected bacteria can produce phages and lyses under appropriate environmental conditions. This relationship between phage and its host is called lysogeny (Figure 12.19).

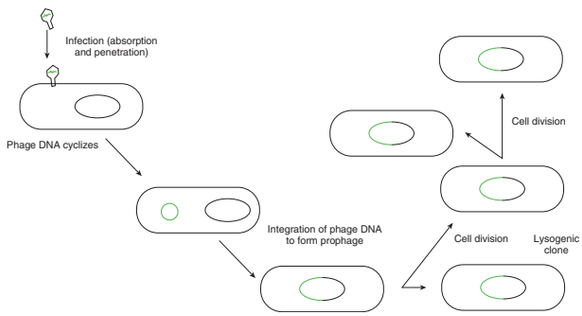


Figure 12.19: Lysogeny

Bacteria that can produce phage particles under some conditions are said to be **lysogens** or **lysogenic bacteria**. Phages which are able to establish lysogeny are called **temperate phages**.

The latent form of virus genome that remains within the host without destroying the host is called the prophage.

The prophage usually is integrated into the bacterial genome. Sometimes phage reproduction is triggered in a lysogenized culture by exposure to UV radiation or other factors. The lysosomes are then destroyed and new phages released – This phenomenon is called induction (Figure 12.20).

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